TELCO | CUSTOMER CHURN ANALYSIS

Executive Overview — Historical Customer Analysis

Overview | Billing | Insights | Diagnostics

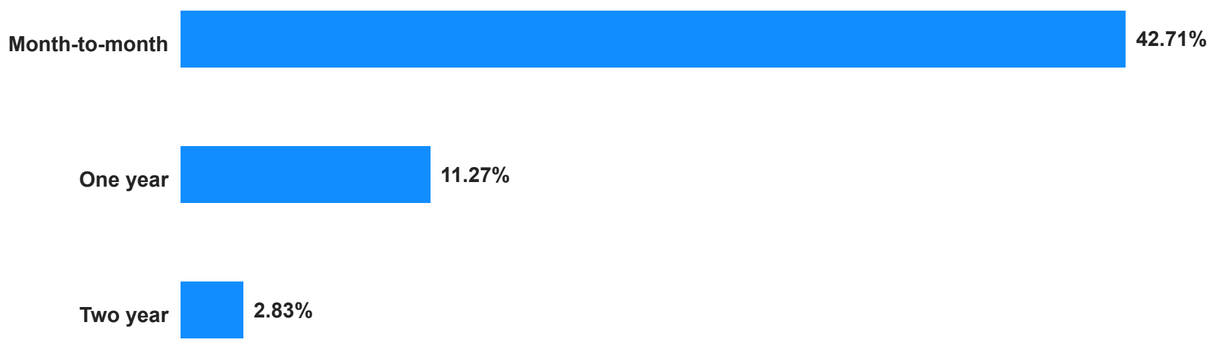
Customers
7043
Unique customer records

Churned
1869
Recorded Cancellations

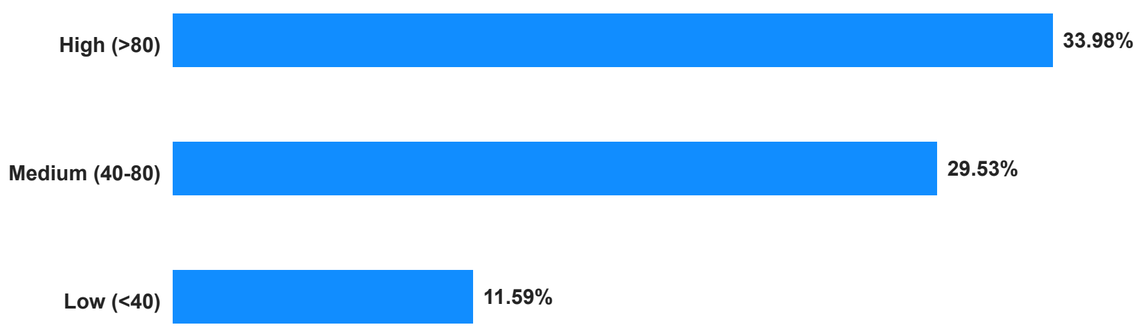
Observed Churn
26.54%
Historical population share

Monthly Billing Exposure
139.1K
Monthly charges - Not verified loss

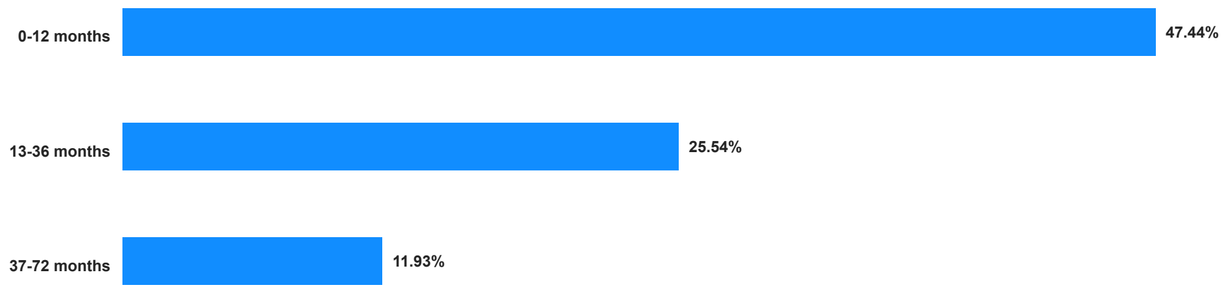
Observed Churn by Contract Type



Observed Churn by Monthly Charge Band



Observed Churn by Customer Tenure



Monthly | One year | Two year

MANAGEMENT SUMMARY

Overall historical findings — Full dataset (7,043 customers)

Month-to-month customers have the highest observed churn (42.71%), compared with 2.83% for two-year contracts. Churn is also elevated among customers in their first 12 months (47.44%).

Recommended next step: Investigate early-tenure customer experience and the drivers of cancellation before designing a targeted retention pilot.

Historical associations only; causation and financial benefits have not been established.

TELCO | CUSTOMER CHURN ANALYSIS

Billing Exposure - Historical Monthly Charges

Overview | **Billing** | Insights | Diagnostics

Churn-Associated Monthly Billing
139,130.85

Month-to-month Share
86.86%

Largest segment share
50.98%

Billing Exposure by Customer Segment

Contract	Low	Medium	High	Total
Month-to-month	3.56%	32.32%	50.98%	**86.86%**
One year	0.20%	2.04%	7.91%	**10.15%**
Two year	0.08%	0.45%	2.46%	**2.99%**
Total	**3.84%**	**34.81%**	**61.34%**	**100.00%**

Share of Churn-Associated Monthly Billing by Contract and Charge Band



Priority Customer Segment

73.49%

Monthly . High charge . 0-12 months

123 historical non-churned records for eligibility review

MANAGEMENT TAKEAWAY

Month-to-month contracts account for **86.86%** of historical churn-associated monthly billing. The month-to-month, high-charge segment represents **50.98%** of overall exposure.

Recommendation: Investigate early-tenure, high-charge customers and verify the eligibility of the **123 historical non-churned records** before considering a targeted retention pilot.

Historical billing exposure does not represent confirmed revenue loss. No retention intervention has been implemented.

TELCO | CUSTOMER CHURN ANALYSIS

Management Insights - Statistical Evidence & Recommendations

Overview | Billing | **Insights** | Diagnostics

Highest Contract Risk	Highest Tenure Risk	Highest Charge Risk
42.71%	47.44%	33.98%

Statistical Evidence - Hypothesis Testing

H1 — Contract Type — $p < 0.001$
Chi-square test . Cramér's V = 0.4101

H2 — Customer Tenure — $p < 0.001$
Mann–Whitney U test . r = -0.4797

H3 — Monthly Charges — $p < 0.001$
Mann-Whitney U test . r = +0.2416

Adjusted Odds Ratios

Adjusted odds ratios indicate associations, not changes in churn probability or causation.

Month-to-month vs Two year	609.67% higher estimated odds	7.10 x
One year vs Two year	135.73% higher estimated odds	2.36x
Tenure (+12 months)	34.21% lower estimated odds	0.66x
Monthly charges (+10 units)	33.69% higher estimated odds	1.34x

PROPOSED MANAGEMENT ACTIONS

01 Investigate
Validate the currently active target population

02 PILOT
Run proactive service check-ins against a comparison group.

03 Decide
Assess churn reduction and costs before scaling

Analytical limitation: *No intervention has been implemented; effectiveness and commercial benefits remain unproven.*

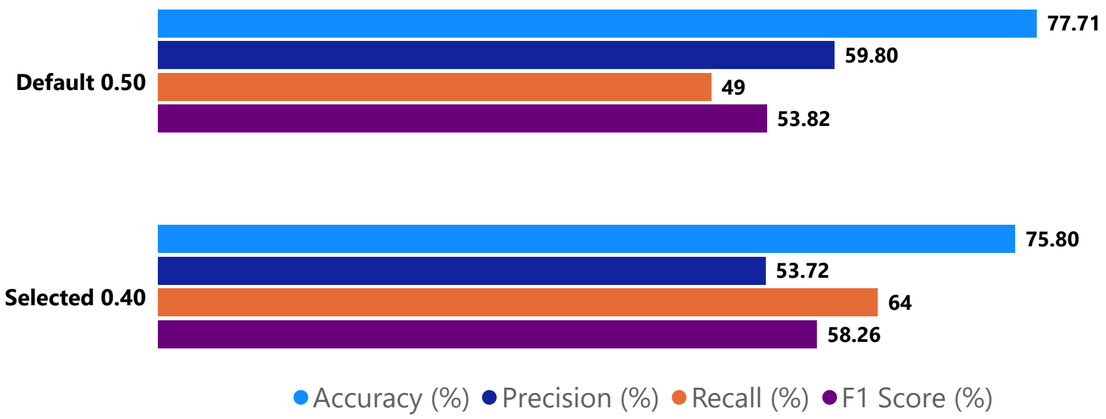
TELCO | CUSTOMER CHURN ANALYSIS

Model Diagnostics - Predictive Performance & Validation

Overview | Billing | Insights | **Diagnostics**

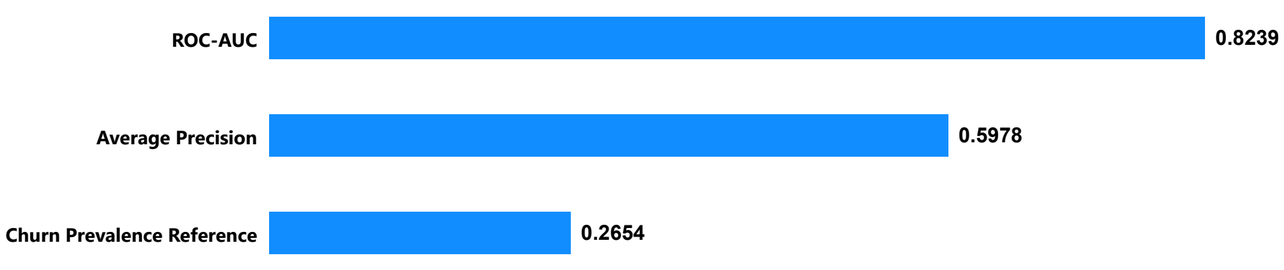
Accuracy	Precision	Recall	F1 Score
75.80%	53.72%	63.64%	58.26%

Model Performance — Threshold Comparison



Logistic Regression — Ranking Performance

Average Precision is compared with the churn prevalence reference (0.2654). ROC-AUC has a different chance-level reference of 0.5.



Confusion Matrix — Logistic Regression (Threshold 0.40)

Actual vs Predicted Churn | Held-out Test Sample: 1,409 Customers

Actual Churn	No	Yes	Total
No	830	205	**1035**
Yes	136	238	**374**
Total	**966**	**443**	**1409**

MANAGEMENT INTERPRETATION

At threshold **0.40**, the model identifies **55 more historical churners** than at 0.50, but generates **82 additional false positives**.

Recommendation: Evaluate the costs of missed churners versus unnecessary interventions before selecting a threshold for operational use.

MODEL LIMITATIONS

The model was evaluated on historical held-out data using contract type, tenure and monthly charges. Future-period performance, probability calibration and commercial impact remain unvalidated.

The model has not been deployed, and no retention intervention or financial benefit has been demonstrated.